THE CANONICAL DELTA-FUNCTOR STRUCTURE ON THE RIGHT DERIVED FUNCTORS

THE STACKS PROJECT AUTHORS

Editorial note. Counted span is the author's own proof environment, source lines 5913-5931, following the statement at 5897-5911.

Editorial note. No omissions: both parts assemble named complete lemmas with the connecting steps identified.

Allowed standard prerequisites (editorial).

- Canonical triangle of a short exact sequence. lemma-derived-canonical-delta-functor (same chapter, cited at ~5922). Comp^+(A) -> D^+(A) is a delta-functor.
- Composition lemmas for delta-functors. lemma-compose-delta-functor-homological (carded L26228) and lemma-exact-compose-delta-functor (same chapter, cited at ~5926). Transfer the delta-functor structure through RF and H^0.
- Effaceability implies universality. Homology, Lemma homology-lemma-efface-implies-universal (cited at ~5929) and lemma-F-acyclic (same chapter, cited at ~5930). Part (2): right acyclic objects efface the functors R^iF for i > 0.

Source reference. Original definition 05RU: sources/stacks-derived/pinned-original-source.tex, source lines 3907–3907. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 010Q: focused-seeds/geometry/S9-der-right-derived-delta-functor/ref-tags.txt, source lines 5–5. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 010S: focused-seeds/geometry/S9-der-right-derived-delta-functor/ref-tags.txt, source lines 6–6. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original reference 010T: focused-seeds/geometry/S9-der-right-derived-delta-functor/ref-tags.txt, source lines 7–7. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Date: stacks-project:derived.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 015C: sources/stacks-derived/pinned-original-source.tex, source lines 5836–5836. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 05SR: sources/stacks-derived/pinned-original-source.tex, source lines 1206–1206. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0152: sources/stacks-derived/pinned-original-source.tex, source lines 4182–4182. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Source reference. Original lemma 0151: sources/stacks-derived/pinned-original-source.tex, source lines 1187–1187. Navigation pointer only: the counted proof cites this statement by \ref as a standing input; the deduction the counted proof has to make is carried out inside the counted span, and no part of that deduction is delegated to this pointer.

Lemma 0.1. *Let $F : \mathcal{A} \to \mathcal{B}$ be an additive functor between abelian categories and assume $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is everywhere defined.*

1. *The functors R^iF, $i \geq 0$ come equipped with a canonical structure of a δ-functor from $\mathcal{A} \to \mathcal{B}$, see Homology, Definition 010Q.*
2. *If every object of $\mathcal{A}$ is a subobject of a right acyclic object for F, then $\{R^iF, \delta\}_{i\geq 0}$ is a universal δ-functor, see Homology, Definition 010S.*

Proof. The functor $\mathcal{A} \to \mathrm{Comp}^+(\mathcal{A})$, $A \mapsto A[0]$ is exact. The functor $\mathrm{Comp}^+(\mathcal{A}) \to D^+(\mathcal{A})$ is a δ-functor, see Lemma 0152. The functor $RF : D^+(\mathcal{A}) \to D^+(\mathcal{B})$ is exact. Finally, the functor $H^0 : D^+(\mathcal{B}) \to \mathcal{B}$ is a homological functor, see Definition 05RU. Hence we get the structure of a δ-functor from Lemma 05SR and Lemma 0151. Part (2) follows from Homology, Lemma 010T and the description of acyclics in Lemma 015C. □